

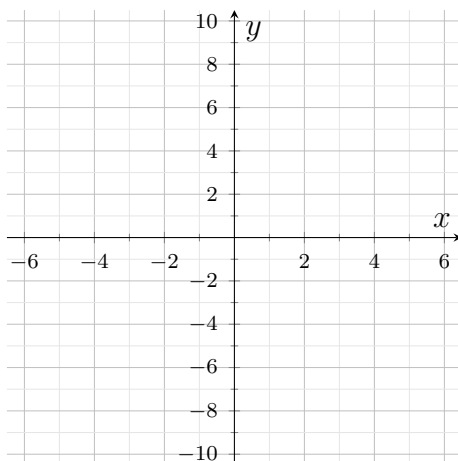
Vertex, Axis, Intercepts and Opening Direction

Reading a parabola from vertex form, from standard form, and from its intercepts and opening direction

Directions. Each grid below is the same size and is blank on purpose — it is your plotting space. For every function, find the features the problem asks for, then sketch the curve on its grid, marking the vertex, drawing the axis of symmetry as a dashed vertical line, and plotting the intercepts you found.

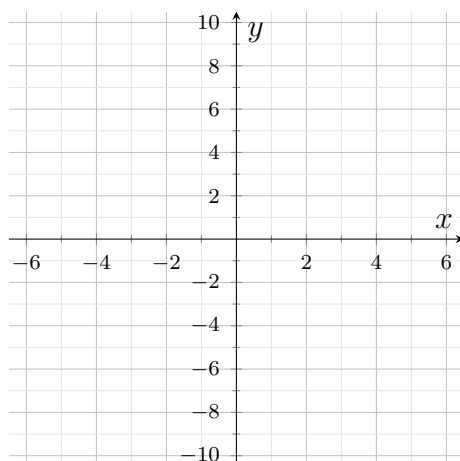
- **Vertex form** $y = a(x - h)^2 + k$: the vertex is (h, k) and the axis of symmetry is $x = h$. Watch the sign — $(x + p)^2$ means h is *negative*.
- **Standard form** $y = ax^2 + bx + c$: the axis of symmetry is $x = -\frac{b}{2a}$; put that x back into the equation for the vertex height. The y -intercept is c .
- **Intercept form** $y = a(x - p)(x - q)$: the x -intercepts are p and q , and the axis of symmetry sits exactly midway between them.
- **Opening direction**: $a > 0$ opens up and the vertex is the lowest point; $a < 0$ opens down and the vertex is the highest point.

1. $y = (x - 3)^2 - 4$. Write the vertex and the axis of symmetry, say which way the parabola opens and how you know, then sketch it.



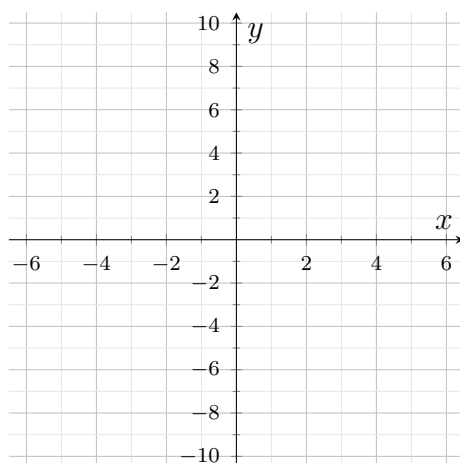
Answer: _____

- 2.** $y = -(x + 2)^2 + 5$. Write the vertex and the axis of symmetry, give the y -intercept, say which way the parabola opens, then sketch it.



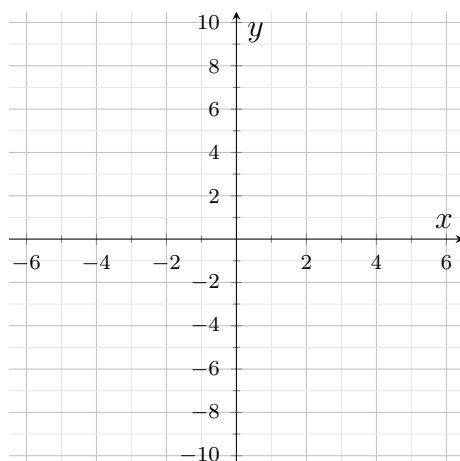
Answer: _____

- 3.** $y = x^2 + 4x - 5$. Use $x = -\frac{b}{2a}$ to find the axis of symmetry, then give the vertex and the y -intercept and sketch the curve.



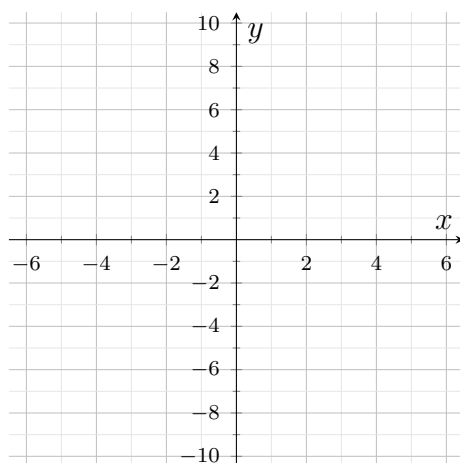
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4. $y = x^2 - 2x - 8$. Factor to get the x -intercepts, then use the fact that the axis of symmetry is midway between them to find the vertex. Sketch the parabola.



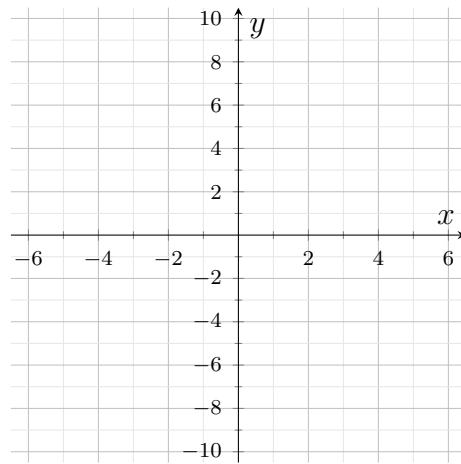
Answer: _____

5. $y = 2(x-1)^2 - 8$. Give the vertex, then solve $2(x-1)^2 - 8 = 0$ for the two x -intercepts. Is this parabola narrower or wider than $y = x^2$? Sketch it.



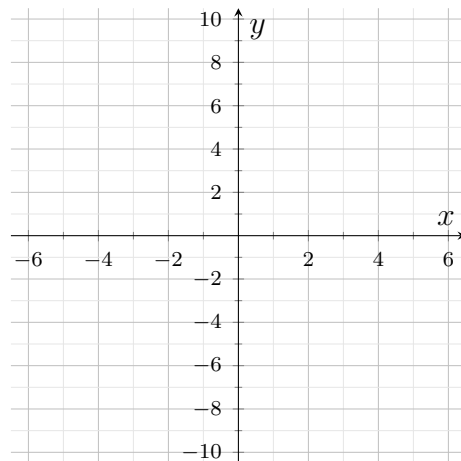
Answer: _____

6. $y = -x^2 + 6x - 5$. Find the axis of symmetry and the vertex, and give the y -intercept. Is the vertex a maximum or a minimum, and why? Sketch the parabola.



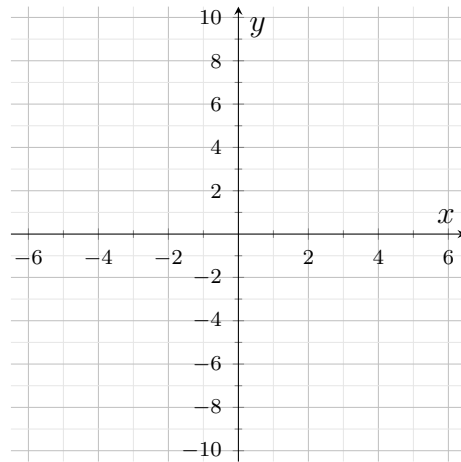
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7. $y = x^2 - 9$. Give the x -intercepts and the vertex. Where is the axis of symmetry, and what is it about this equation that puts it there? Sketch the curve.



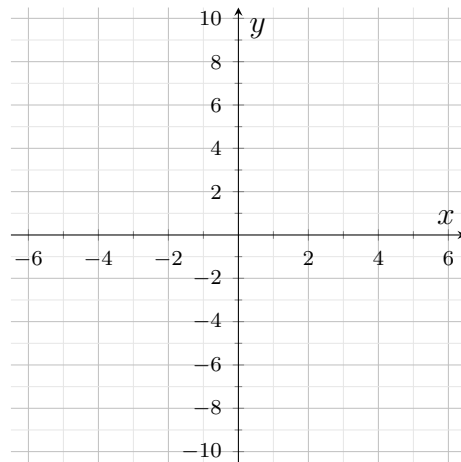
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8. $y = \frac{1}{2}(x + 4)^2 - 2$. Give the vertex and both x -intercepts. Will this parabola be narrower or wider than $y = x^2$, and which way does it open? Sketch it.



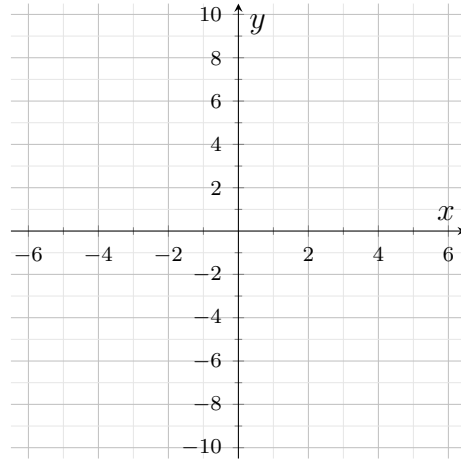
Answer: _____

9. $y = 2x^2 - 12x + 10$. Factor out the leading coefficient first, give the two x -intercepts, then get the vertex from the midpoint of those intercepts. Check your axis against $x = -\frac{b}{2a}$ and sketch.



Answer: _____

10. A parabola crosses the x -axis at -1 and at 5 , and it also passes through the point $(2, 9)$. Write its equation in the form $y = a(x - p)(x - q)$ and find a , then multiply out to standard form. Give the vertex and the opening direction. Finally sketch the curve, labelling the vertex, the axis of symmetry, both x -intercepts and the y -intercept, and write one sentence saying how the extra point decided the opening direction.



Answer: _____